FOUNDATION_DROOLS - Business Rules Engine Foundation

Developer implementation guide: `FOUNDATION_DROOLS_DEV_IMPL_GUIDE.md`.

1. Purpose

This foundation defines how SOPHIX uses Drools/KIE Server to execute configurable business rules outside normal application code.

Use this document when implementing:

- Rule-driven validation.
- Eligibility decisions.
- Pricing or fee applicability rules.
- Dunning escalation rules.
- Workflow precondition checks.
- Operator-specific rule packages.
- KJAR build/deploy/versioning.

2. Ownership

FOUNDATION_DROOLS owns:

- KIE Server deployment topology.
- KJAR packaging conventions.
- Rule invocation contract.
- Fact/result modeling conventions.
- Rule versioning and deployment rules.
- Error handling and monitoring expectations.

FOUNDATION_DROOLS does not own:

- The business meaning of rules. Each DD owns its own rule logic.
- Rule facts for a specific module. The owning module defines them.
- Workflow orchestration. See `FOUNDATION_CAMUNDA.md`.
- API authentication. See `FOUNDATION_AUTH.md`.

3. When To Use Drools

Use Drools when rules are:

- Business-owned or likely to change by operator/country.
- A decision table or many conditional rules.
- Needed consistently by multiple code paths.
- Better audited by rule ID than hidden in imperative code.
- Operator-specific, for example KE WIK vs WUG vs YASSN.

Do not use Drools for:

- Simple field validation.
- Database constraints.
- One-off `if` statements unlikely to vary.

- High-frequency inner loops where a normal code branch is cheaper and clearer.

4. Runtime Architecture

```
flowchart LR
    JAVA[Java modules] -->|HTTPS JSON| LB[NGINX or LB]
    PHP[Laravel modules] -->|HTTPS JSON| LB
    LB --> K1[KIE Server node 1]
    LB --> K2[KIE Server node 2]
    K1 -.load KJAR.-> REPO[(Maven / artifact repo)]
    K2 -.load KJAR.-> REPO
```

KIE Server is stateless for SOPHIX calls. Each request sends facts in and receives rule results out. The caller owns persistence and business action.

5. Deployment Topology

Item	Requirement
Runtime	Drools 8 / KIE Server
Nodes	2 KIE Server nodes
Front door	NGINX or load balancer
State	Stateless request/response
Rule artifacts	KJARs loaded from Maven/artifact repository
Auth	Basic auth or internal service auth over TLS
Endpoint	`https://kie.sophix.internal/kie-server/services/rest/server`

If one KIE node fails, the load balancer routes to the surviving node. There is no per-request session state to recover.

6. KJAR Layout

Each module that owns rules owns a rules artifact.

Recommended layout:

```
modules/{module}-rules/
■■■ pom.xml
■■■ src/main/resources/
■   ■■■ META-INF/kmodule.xml
■   ■■■ rules/
■       ■■■ {dd-code}-{purpose}.drl
■       ■■■ {dd-code}-{operator}-{purpose}.drl
■■■ src/main/java/com/sophix/{module}/facts/
    ■■■ RequestFact.java
    ■■■ ContextFact.java
    ■■■ DecisionResult.java
    ■■■ ValidationError.java
```

`pom.xml` packaging:

```
<packaging>kjar</packaging>
```

`kmodule.xml`:

```
<kmodule xmlns="http://www.drools.org/xsd/kmodule">
  <kbase name="subscription-rules" packages="com.sophix.subscription.rules">
    <ksession name="subscription-session" default="true"/>
  </kbase>
</kmodule>
```

Rule file example:

```
package com.sophix.subscription.rules

import com.sophix.subscription.facts.ActivationRequestFact
import com.sophix.subscription.facts.ValidationError

rule "R-SUB-ACT-001 subscription must be CREATED"
when
    $req: ActivationRequestFact(statusCode != "CREATED")
then
    insert(new ValidationError(
        "R-SUB-ACT-001",
```

```
            "statusCode",
            "Subscription must be CREATED before activation"
        ));
end
```

7. Laravel-Owned Rules

Laravel modules may own KJARs without writing Java application code. Define fact types in DRL with `declare`.

```
package com.sophix.fulfillment.rules

declare PaymentFact
    amount   : java.math.BigDecimal
    expected : java.math.BigDecimal
end

declare PaymentDecision
    classification : String
    amountShort    : java.math.BigDecimal
end

rule "R-FUL-PAY-001 classify partial payment"
when
    $p: PaymentFact(amount < expected)
then
    insert(new PaymentDecision("PARTIAL", $p.getExpected().subtract($p.getAmount())));
end
```

This keeps the HTTP contract the same for PHP and Java callers.

8. Container Naming and Versioning

Container ID pattern:

```
{module}-rules-{operator}_{version}
```

Examples:

```
subscription-rules-wik_1.0.0
billing-rules-wik_1.2.0
workorder-rules-wik_1.0.0
```

Versioning rules:

Rule	Description
DROOLS-VER-1	Version KJARs semantically: MAJOR.MINOR.PATCH.
DROOLS-VER-2	Patch version for rule bug fix that preserves facts/results.
DROOLS-VER-3	Minor version for additive facts/results or new rules.
DROOLS-VER-4	Major version for breaking fact/result changes.
DROOLS-VER-5	Caller config must pin the container ID. Do not silently float production callers to latest.
DROOLS-VER-6	Keep older KJAR versions available while workflows that started on them may still run.

9. HTTP Invocation Contract

Endpoint:

```
POST /kie-server/services/rest/server/containers/instances/{containerId}
Authorization: Basic <internal>
Content-Type: application/json
X-KIE-ContentType: JSON
```

Request:

```
{
  "lookup": "subscription-session",
  "commands": [
    {
      "insert": {
        "object": {
          "com.sophix.subscription.facts.ActivationRequestFact": {
            "subscriptionId": "sub_01HX",
            "statusCode": "CREATED",
```

```
              "operatorCode": "WIK"
            }
          },
          "out-identifier": "request"
        }
      },
      { "fire-all-rules": { "max": -1 } },
      { "get-objects": { "out-identifier": "results" } }
    ]
  }
```

Response:

```
  {
    "type": "SUCCESS",
    "result": {
      "execution-results": {
        "results": {
          "results": {
            "value": []
          }
        }
      }
    }
  }
```

Caller rules:

Rule	Description
DROOLS-CALL-1	Callers must set a request timeout. Recommended default: 3 seconds.
DROOLS-CALL-2	Callers must log `containerId`, `lookup`, `correlationId`, and rule result summary.
DROOLS-CALL-3	Do not log sensitive full fact payloads unless redacted.
DROOLS-CALL-4	Treat KIE unavailable as a dependency failure unless the owning DD defines a fallback.
DROOLS-CALL-5	Map validation failures to `BUSINESS_RULE_ERROR` with rule ID.

10. Standard Result Types

Use explicit result facts.

Validation error:

```
  {
    "com.sophix.common.rules.ValidationError": {
      "ruleId": "R-SUB-ACT-001",
      "field": "statusCode",
      "message": "Subscription must be CREATED before activation"
    }
  }
```

Decision:

```
  {
    "com.sophix.common.rules.DecisionResult": {
      "decisionCode": "PAY_FIRST_REQUIRED",
      "ruleId": "R-BIL-001",
      "attributes": {
        "amount": "1500.00",
        "currency": "KES"
      }
    }
  }
```

Rules:

Rule	Description
DROOLS-RES-1	Every failure/decision result must include `ruleId`.
DROOLS-RES-2	Result classes must be stable within a KJAR major version.
DROOLS-RES-3	Multiple validation errors may be returned in one call.
DROOLS-RES-4	Caller decides whether errors are blocking or advisory based on DD rules.

11. Java Caller Example

```
Map<String, Object> body = Map.of(
    "lookup", "subscription-session",
    "commands", List.of(
        Map.of("insert", Map.of(
            "object", Map.of(
                "com.sophix.subscription.facts.ActivationRequestFact",
                Map.of(
                    "subscriptionId", "sub_01HX",
                    "statusCode", "CREATED",
                    "operatorCode", "WIK"
                )
            ),
            "out-identifier", "request"
        )),
        Map.of("fire-all-rules", Map.of("max", -1)),
        Map.of("get-objects", Map.of("out-identifier", "results"))
    )
);

HttpRequest req = HttpRequest.newBuilder()
    .uri(URI.create(kieBaseUrl + "/containers/instances/subscription-rules-wik_1.0.0"))
    .header("Content-Type", "application/json")
    .header("X-KIE-ContentType", "JSON")
    .header("Authorization", basicAuth)
    .timeout(Duration.ofSeconds(3))
    .POST(HttpRequest.BodyPublishers.ofString(json.writeValueAsString(body)))
    .build();
```

12. Laravel Caller Example

```
$response = Http::withBasicAuth(config('rules.user'), config('rules.password'))
    ->timeout(3)
    ->withHeaders([
        'Content-Type' => 'application/json',
        'X-KIE-ContentType' => 'JSON',
    ])
    ->post(config('rules.base_url') . '/containers/instances/fulfillment-rules-wik_1.0.0', [
        'lookup' => 'fulfillment-session',
        'commands' => [
            [
                'insert' => [
                    'object' => [
                        'com.sophix.fulfillment.rules.PaymentFact' => [
                            'amount' => '50000.00',
                            'expected' => '60000.00',
                        ],
                    ],
                    'out-identifier' => 'payment',
                ],
            ],
            ['fire-all-rules' => ['max' => -1]],
            ['get-objects' => ['out-identifier' => 'results']],
        ],
    ]);

$results = $response->json('result.execution-results.results.results.value', []);
```

13. Error Handling

Condition	Caller behavior
KIE HTTP timeout	Return dependency failure or retry according to owning DD.
KIE returns non-2xx	Treat as dependency failure.
KIE response type not SUCCESS	Treat as rule-engine failure.
ValidationError result	Return `BUSINESS_RULE_ERROR` or route workflow to rejection path.
Empty results	Usually means no rule rejected and no positive decision needed. Confirm with owning DD.
Unknown result class	Treat as integration error; alert.

Recommended API error:

```
{
  "error": {
```

```
    "code": "BUSINESS_RULE_ERROR",
    "message": "Activation request failed rule validation.",
    "ruleId": "R-SUB-ACT-001",
    "field": "statusCode",
    "correlationId": "corr_01HX"
  }
}
```

14. Testing

Required tests:

Test type	Requirement
Rule unit tests	Each DRL package has test facts for pass/fail paths.
Contract tests	Caller verifies expected result classes and JSON paths.
Integration tests	Staging caller hits staging KIE Server container.
Regression tests	New KJAR version runs old fixture set before deploy.
Performance smoke	High-volume rules checked for acceptable latency.

15. Deployment Flow

1. Developer updates DRL/facts.
2. CI runs rule unit tests.
3. CI builds KJAR.
4. CI publishes KJAR to artifact repository.
5. KIE Server loads or reloads the container version.
6. Staging callers run integration tests.
7. Production caller config is updated to the new container ID.
8. Monitor rule latency and error rate after rollout.

Production rule: do not overwrite an existing KJAR version. Publish a new version.

16. Monitoring

Alert on:

Signal	Threshold
KIE node count	Less than 2
Rule invocation p95 latency	Above module-specific threshold
KIE HTTP 5xx	Any sustained increase
KIE timeout rate	Any sustained increase
Container load failures	Any
Rule result integration errors	Any
Artifact repository unavailable	Any during deploy window

17. Medium Budget Infrastructure

17.1 Server Dimensions

Required medium baseline:

Server role	Count	CPU	RAM	Disk	Notes
KIE Server node	2	2-4 vCPU each	8 GB each	80-100 GB SSD each	Stateless app nodes.
Artifact repository	Shared platform service	2 vCPU	4-8 GB	100 GB+	Maven/Nexus/Artifactory or equivalent.
Load balancer	1 minimum, 2 preferred	1-2 vCPU	2 GB	20 GB SSD	NGINX/LB if on-prem.

Server role	Count	CPU	RAM	Disk	Notes
Monitoring/log storage	Shared platform service	2 vCPU	4 GB	Depends on retention	Stores KIE logs and metrics.

Minimum acceptable start:

```
2 x KIE nodes: 2 vCPU / 8 GB RAM / 80 GB SSD
1 x LB node:   1-2 vCPU / 2 GB RAM / 20 GB SSD
```

Preferred medium:

```
2 x KIE nodes: 4 vCPU / 8-16 GB RAM / 100 GB SSD
2 x LB nodes:  1-2 vCPU / 2 GB RAM / 20 GB SSD
```

Placement rules:

- Do not place both KIE nodes on the same physical host.
- Do not co-locate KIE nodes with PostgreSQL primaries if avoidable.
- Keep KIE endpoint private to SOPHIX modules.
- Artifact repository should be backed up; rules are deployable artifacts.

17.2 Recommended OS

Use one operator-standard Linux distribution:

OS	Recommendation
Ubuntu Server 22.04 LTS or 24.04 LTS	Recommended default for medium teams.
Rocky Linux 9 / RHEL 9	Use if the operator standardizes on RHEL-family systems.

OS baseline:

- 64-bit minimal server install.
- No desktop packages.
- `chrony` or equivalent NTP enabled.
- Security patching through operator maintenance process.
- SSH restricted to admin VPN/network.
- Host firewall enabled.
- Java runtime pinned and tested with selected Drools/KIE version.

17.3 Disk Partitions

For each KIE node with 80-100 GB disk:

```
/             30 GB
/opt/kie      20 GB
/var/log      20 GB
/tmp          10 GB
free/reserved remaining
```

Partition rules:

- KIE nodes should not store durable business state.
- Logs must not fill `/`.
- KJARs are cached locally but source of truth is the artifact repository.

17.4 JVM Starting Point

For 8 GB RAM nodes:

```
JAVA_OPTS="-Xms2g -Xmx4g"
```

For 16 GB RAM nodes:

```
JAVA_OPTS="-Xms4g -Xmx8g"
```

Tune after observing rule latency, GC, and heap usage.

17.5 Network Ports

Flow	Port	Exposure
Module to KIE LB	443	Internal application network only
LB to KIE node	8080 or 8443	Internal only
KIE node to artifact repository	443 or repository port	Internal only
Admin SSH	22	Admin VPN/network only
Monitoring scrape	Operator standard	Monitoring network only

18. AWS Deployment Recommendations

This section applies only when deploying KIE Server on AWS.

18.1 Recommended AWS Shape

Component	AWS service	Recommendation
KIE runtime	ECS on Fargate, ECS on EC2, or EC2 Auto Scaling Group	ECS/Fargate is preferred if the team already uses containers. EC2 ASG is simpler for VM-based teams.
Load balancer	Internal Application Load Balancer	Keep KIE private to SOPHIX services.
Artifact storage	CodeArtifact, S3-backed repo, Nexus/Artifactory on EC2/ECS	Must support Maven artifacts.
Shared file storage	EFS only if repository/runtime requires shared files	KIE itself should load artifacts from repository, not depend on shared disk.
Secrets	AWS Secrets Manager or SSM Parameter Store	Store KIE credentials and repository credentials.
Logs	CloudWatch Logs	Ship KIE logs.

18.2 Compute Sizing

ECS/Fargate medium start:

```
Desired tasks: 2
Task size:     2 vCPU / 8 GB memory
Placement:     Spread across 2 Availability Zones
```

EC2 ASG medium start:

```
2 x EC2 instances: m7g.large or m6i.large
EBS:               80-100 GB gp3 each
```

Preferred medium:

```
2 x tasks/nodes: 4 vCPU / 8-16 GB memory
```

Use Graviton only after verifying the chosen KIE/Drools image and JDK work cleanly on ARM. Otherwise use x86 instances.

18.3 AWS Network Layout

Recommended VPC layout:

```
Private app subnets:
  - KIE tasks/instances
  - SOPHIX module services

Private artifact subnet/service:
  - Artifact repository if self-hosted

Public subnets:
  - No KIE resources
```

Security group rules:

Source	Destination	Port	Purpose
Module service SG	Internal ALB SG	443	Rule invocation
ALB SG	KIE task/EC2 SG	8080 or 8443	Forward traffic
KIE SG	Artifact repo SG/service endpoint	443 or repo port	Pull KJARs
Admin VPN/bastion/SSM	KIE nodes	22 or SSM	Admin access if EC2

18.4 AWS Operational Notes

- Use an internal ALB, not public.
- Use ECS task health checks or ALB health checks.
- Store rule artifacts in a Maven-compatible repository.
- Roll out KJAR versions through CI/CD, not by manual copy.
- Use CloudWatch alarms for unhealthy tasks, 5xx rate, latency, and task restarts.
- If using Fargate, choose valid CPU/memory combinations and size at the task level.

Reference docs:

- [AWS Fargate task CPU and memory](https://docs.aws.amazon.com/en_us/AmazonECS/latest/developerguide/fargate-tasks-services.html)
- [Use an Application Load Balancer for Amazon ECS](https://docs.aws.amazon.com/AmazonECS/latest/developerguide/alb.html)
- [Amazon EFS documentation](https://aws.amazon.com/documentation-overview/efs/)

19. Developer Checklist

- [ ] Rule is justified as configurable/business-owned logic.
- [ ] KJAR has stable container ID and version.
- [ ] Rule names include rule IDs.
- [ ] Facts/results are versioned and contract-tested.
- [ ] Caller pins container version.
- [ ] Caller timeout is configured.
- [ ] Sensitive facts are not logged raw.
- [ ] Validation errors map to `BUSINESS_RULE_ERROR`.
- [ ] Integration tests call staging KIE Server.
- [ ] New KJAR version is published; old version is not overwritten.